Lit Modelle

Montag, 2. März 2026 16:00

Two Aspects:

1. Score prediction:

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(y_i - \hat{y}_i)^2}$$

2. Prob prediction:

$$RPS = \frac{1}{r-1}\sum_{i=1}^{r-1}\left(\sum_{j=1}^{i}(p_j - a_j)\right)^2,$$

[2023 Soccer Prediction Challenge](#)

2017 catboost best ([2403.07669](#))

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og for 1. Dixon Coles (1997). From

2307.02139v1

2.1 The Dixon and Coles model

Dixon and Coles (1997) defined a bivariate model with Poisson marginal distributions. The corresponding joint probability mass function (pmf) is given by

$$P(X_1 = x_1, X_2 = x_2) = \tau_{\lambda_1,\lambda_2}(x_1, x_2)\frac{\lambda_1^{x_1}\exp(-\lambda_1)}{x_1!}\frac{\lambda_2^{x_2}\exp(-\lambda_2)}{x_2!},$$

with

$$\tau_{\lambda_1,\lambda_2}(x_1, x_2) = \begin{cases} 1 - \lambda_1\lambda_2\tilde{\omega}, & \text{if } x_1 = x_2 = 0, \\ 1 + \lambda_1\tilde{\omega} & \text{if } x_1 = 0,\ x_2 = 1, \\ 1 + \lambda_2\tilde{\omega} & \text{if } x_1 = 1,\ x_2 = 0, \\ 1 - \tilde{\omega} & \text{if } x_1 = x_2 = 1, \\ 1 & \text{otherwise}, \end{cases}$$

where λ_1 and λ_2 are the means of the two Poisson marginal distributions and $\tau_{\lambda_1,\lambda_2}(\cdot,\cdot)$ measures the correlation between the scores. For $x_1 > 1$ and $x_2 > 1$, the probabilities remain unchanged from the product of the marginal distributions as probabilities are shifted only between the four pairs $(0,0)$, $(1,0)$, $(0,1)$ and $(1,1)$. The magnitude of shifting depends on the dependence parameter $\tilde{\omega}$, which has to satisfy the following inequality:

$$\max(-1/\lambda_1, -1/\lambda_2) \leq \tilde{\omega} \leq \min(1/\lambda_1\lambda_2, 1).$$

Thus, $\tilde{\omega}$ can take positive or negative values but it is of limited range for reasonable means λ_1 and λ_2. The case $\tilde{\omega} = 0$ corresponds to scores being independent.

Evaluating Soccer Match Prediction Models: A Deep Learning Approach and Feature Optimization for Gradient-Boosted Tree

Berrar et al. - 2019 - I...

Yeung et al. - 2023 - E...

1. Berrar Ratings (with/without XGBoost)
2. Recency feature extraction + Omception block + ecoding thouth Transformer and decoding thorugh MLP

Based on these four performance rating features (per team), Eqs. 1 and 2 define a *goal-prediction model* that predicts the *goals* scored by the home and away team, respectively.

$$\hat{g}_h(H_{hatt}, A_{adef}) = \frac{\alpha_h}{1 + \exp(-\beta_h(H_{hatt} + A_{adef}) - \gamma_h)} \quad (1)$$

$$\hat{g}_a(A_{aatt}, H_{hdef}) = \frac{\alpha_a}{1 + \exp(-\beta_a(A_{aatt} + H_{hdef}) - \gamma_a)} \quad (2)$$

A Bayesian approach for predicting match outcomes: FIFA World Cup 2026

JIAMCS2024_1848

2. Bayesian Logisitc Regression (w Offensive score, Fifa rank,...)
2. CatBoost

NESTED ZERO INFLATED GENERALIZED POISSON REGRESSION FOR FIFA WORLD CUP 2022

Gilch - 2022 - Nested Z...

1. Zero inflated poisson (using elo)

Gilch und Müller - 2...

1. Nested Poisson (selben atzen)

Prediction of **the** FIFA World Cup 2018– A random forest approach with an emphasis on estimated team ability parameters

Groll et al. - 2018 - Pr...

1. RF (with odds and FIFA Rank), Poisson & Ranking Methods. Best perfoming was RF with ranking included as feautre

Evaluating Predictors for the 2022 World Cup Using Decision Trees and Random Forests

MorgenN_2024_Final

2: RF outperforms Elo-Sytle rpedcition

- Lit. f. Goal pred., i.e. tippen, wie Tippspiel.
  ↳ XG Boost, Poisson
  ↳ Rech. ob was besser f. either classification oder Gexakte Tore
  ↳ bisher Elo based sim

=> 2 Modelle (einmal simulieren mit Elo) und einmal exakt predict.

=> suchen ob Elo available